

AK-ACD

Ambiguous
Clearance
Detection

Team. 11

**HAVE YOU EVER HESITATED WHILE DRIVING,
EVEN THOUGH NOTHING ACTUALLY HAPPENED?**



WHAT IF... THIS SITUATION

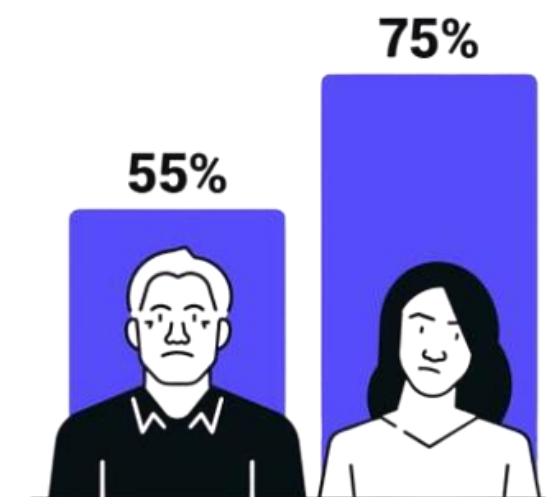


At this situation, most beginner drivers feel **anxiety** highly and need help.

SURVEY ABOUT DRIVING ANXIETY



55% of Americans experience some form of anxiety while performing common driving maneuvers.



Women (75%) reported experiencing driving anxiety **more often than men** (55%).

“ OKAY... SLOW DOWN... DON'T PANIC. ”

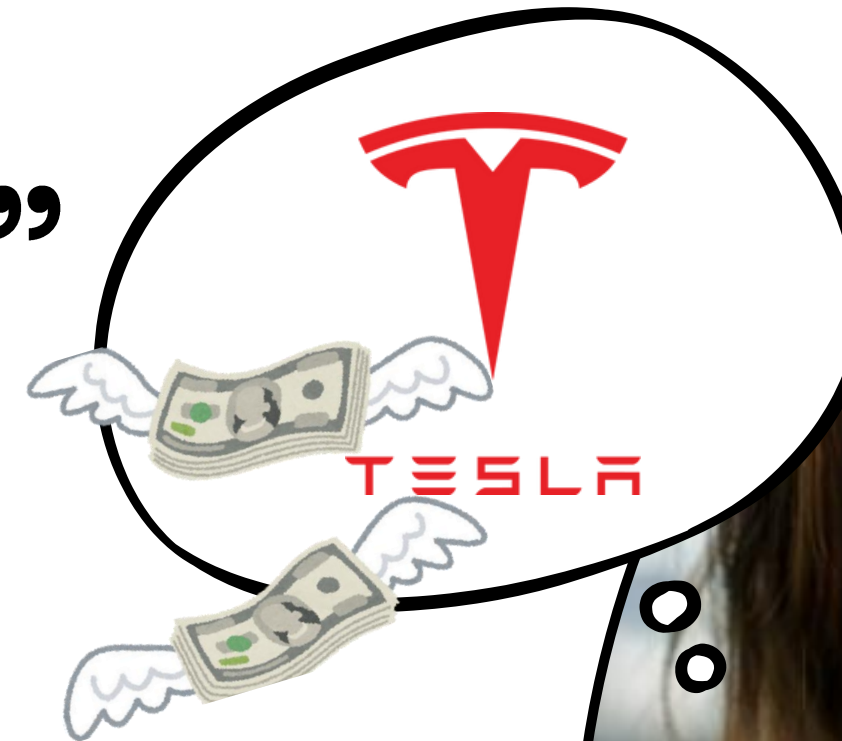
EMILY

Beginner Driver

18-year-old Female

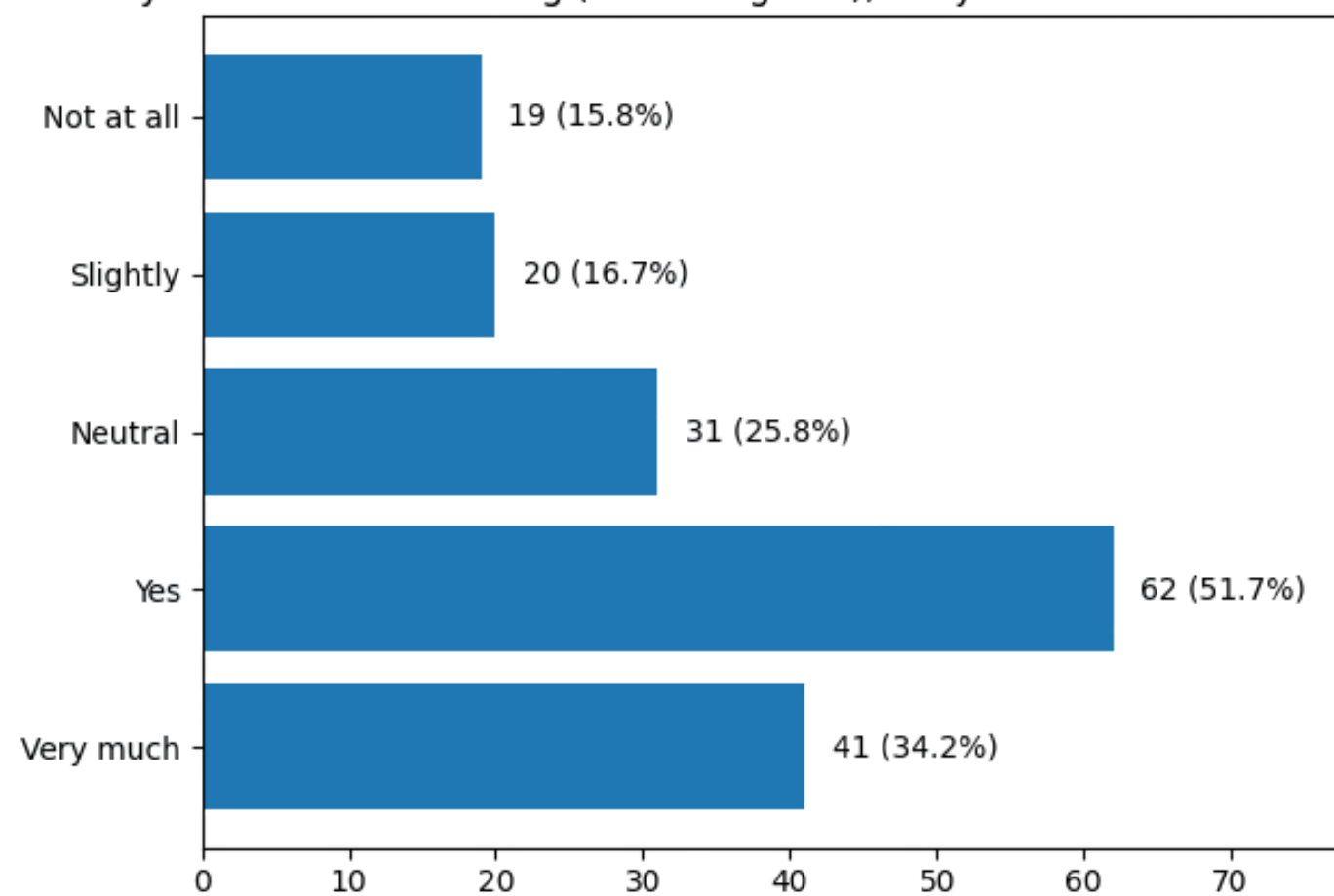
**I always feel nervous when driving,
but I can't afford a car like a Tesla.**

- Anxious about first-time driving experience.
- Lacks confidence in narrow spaces.
- Anxiety increases significantly when visibility is limited.
- Financial difficulties in buying a car with advanced technologies.

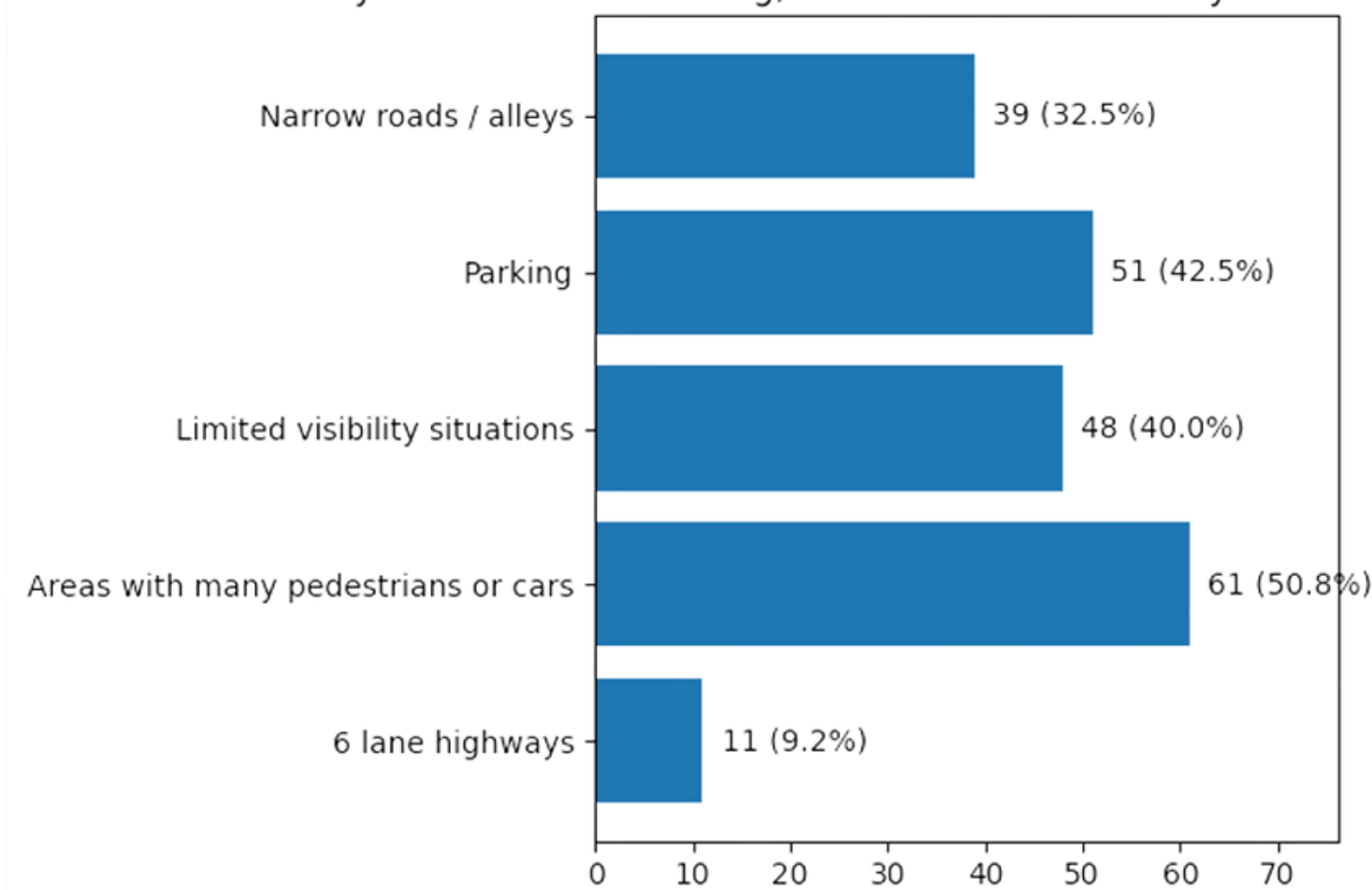




When you first started driving (around age 18), did you feel anxious about driving?



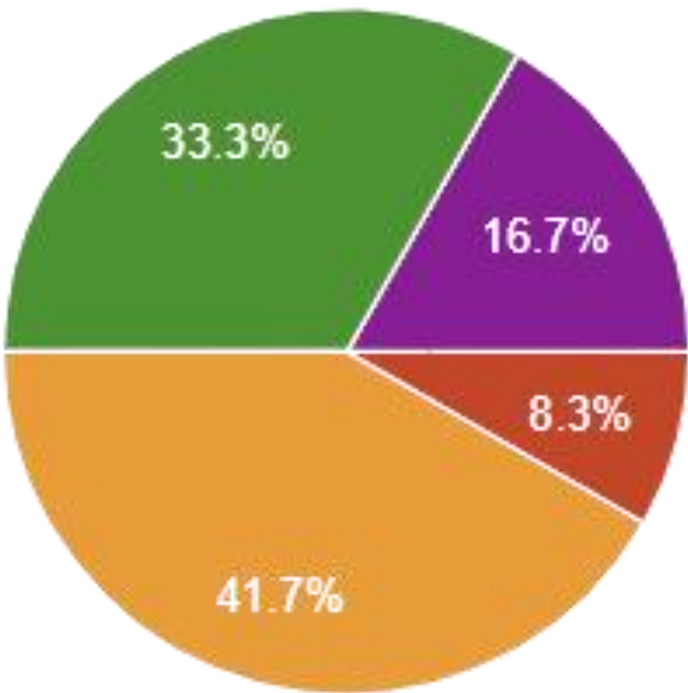
When you first started driving, in which situations did you feel the most anxious?



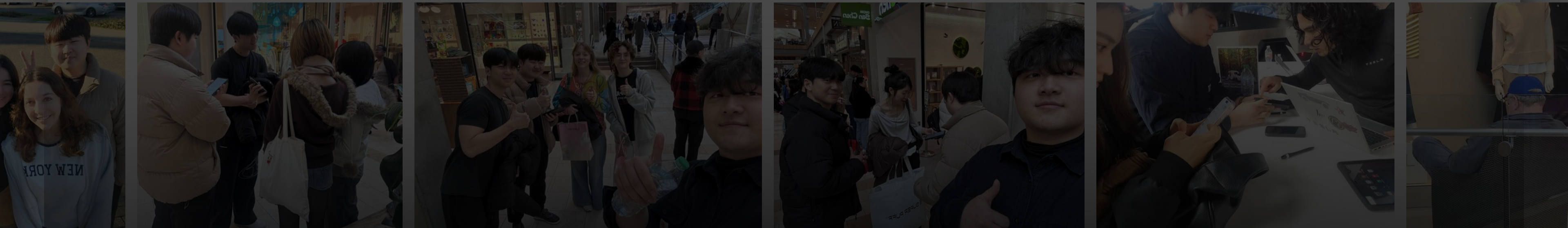


If there were a product that helped you make clear decisions in ambiguous driving situations like the ones above, would you consider using it?

응답 102



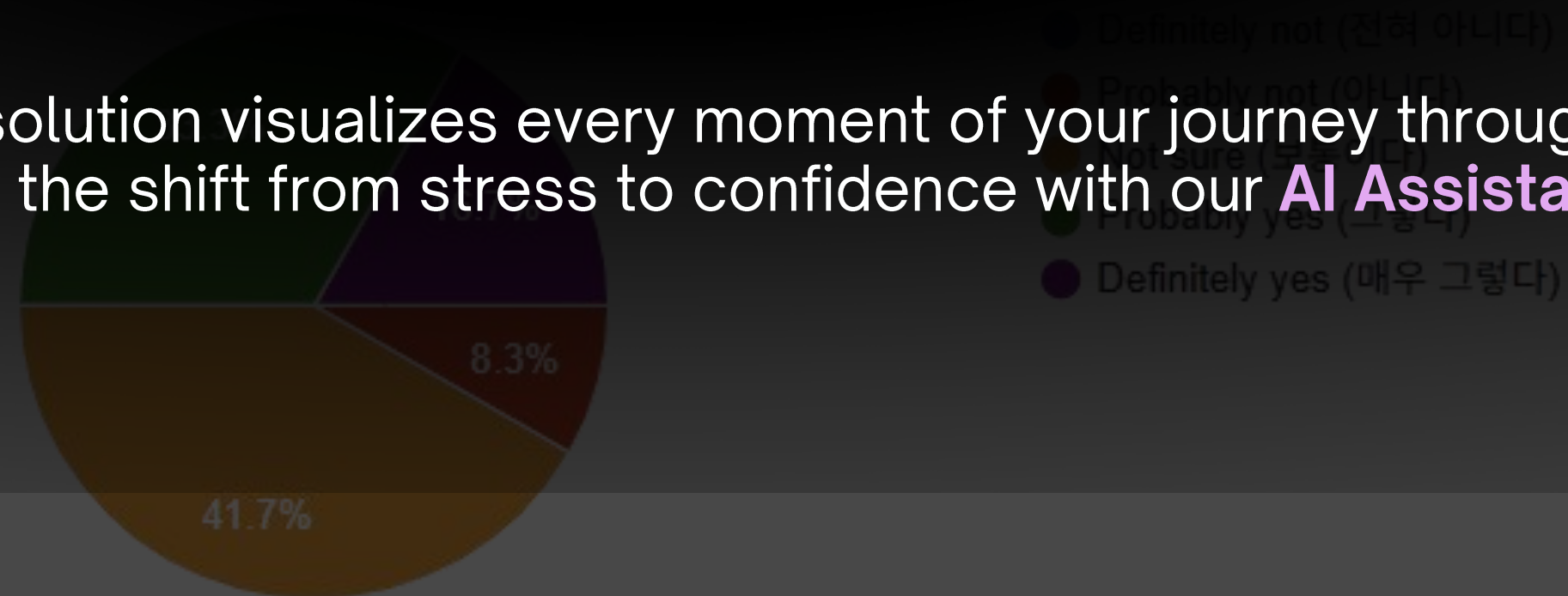
- Definitely not (전혀 아니다)
- Probably not (아니다)
- Not sure (보통이다)
- Probably yes (그렇다)
- Definitely yes (매우 그렇다)



“ASSISTING DRIVERS, NOT AUTOMATING THEM.”

Dissolve the uncertainty of the road with the power of data.

Our intelligent dashcam solution visualizes every moment of your journey through real-time data.
Experience the shift from stress to confidence with our **AI Assistant**.



INSTANT DECISION SUPPORT



- Activate the system in **uncertain** or **anxious driving situations**
- A single camera-based obstacle distance detection
- Warning output when entering a **Risk Zone**

INSTANT DECISION SUPPORT



- Activate the system in **uncertain** or **anxious driving situations**
 - Real-time scanning and data analysis of surrounding hazards
- RED (Stop/Slow):** Immediate warning – Stay/Do Not Enter
- GREEN (Go/Safe):** Safety confirmed – Clear to Proceed

AI MODELS & LIBRARIES



DepthPro



Monocular Metric Depth Estimation

Apple's deep learning model that provides accurate depth information from a single camera input.

Role

Enables accurate depth perception with a standard camera, forming the foundation for all subsequent processing.



SAM2

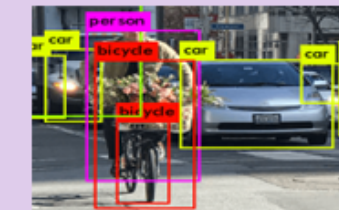


Segmentation-based Extraction

Meta's deep learning model that segments road area and surrounding objects from a single image input

Role

Separates the drivable area and isolates the nearest object, supporting stable clearance analysis in subsequent processing.



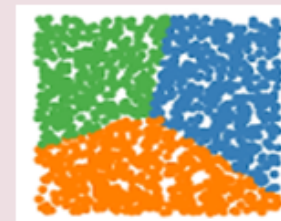
Object & Instance Detection

Detects and classifies objects from a single image input, providing instance-level semantic information.

Role

Provides object identity for downstream spatial analysis, supporting stable clearance estimation.

DBSCAN



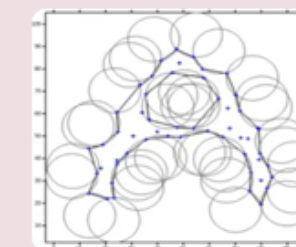
Density-based Clustering

Groups together points that are closely packed together, marking them as part of the same object.

Role

Clusters objects around the vehicle based on their 3D positions in the estimated depth space.

Alpha Shape



Boundary Extraction

Generates a concave boundary (Alpha Shape) of a point set, where the α parameter controls how tightly the boundary fits the data

Role

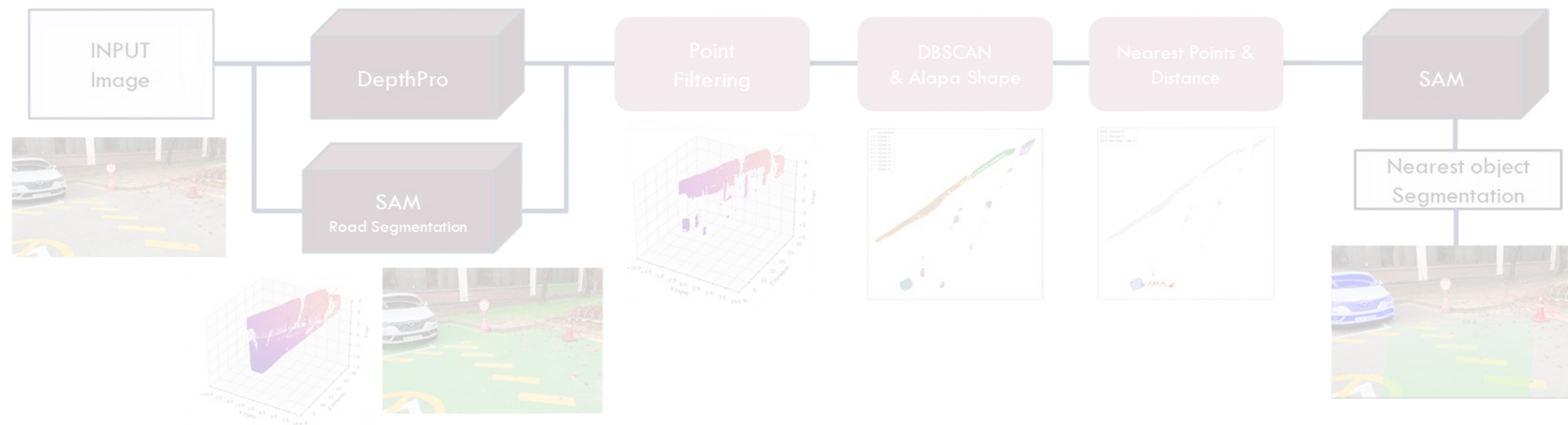
Extracts precise boundaries for each clustered object, essential for accurate distance calculation.

ARCHITECTURE DESIGN

1



2

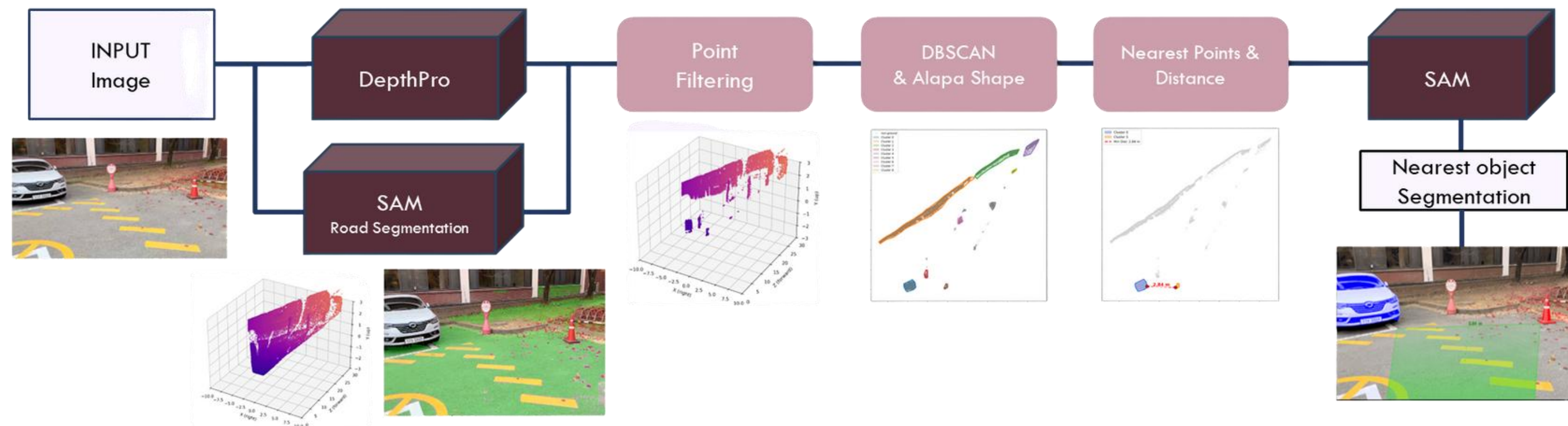


ARCHITECTURE DESIGN

1



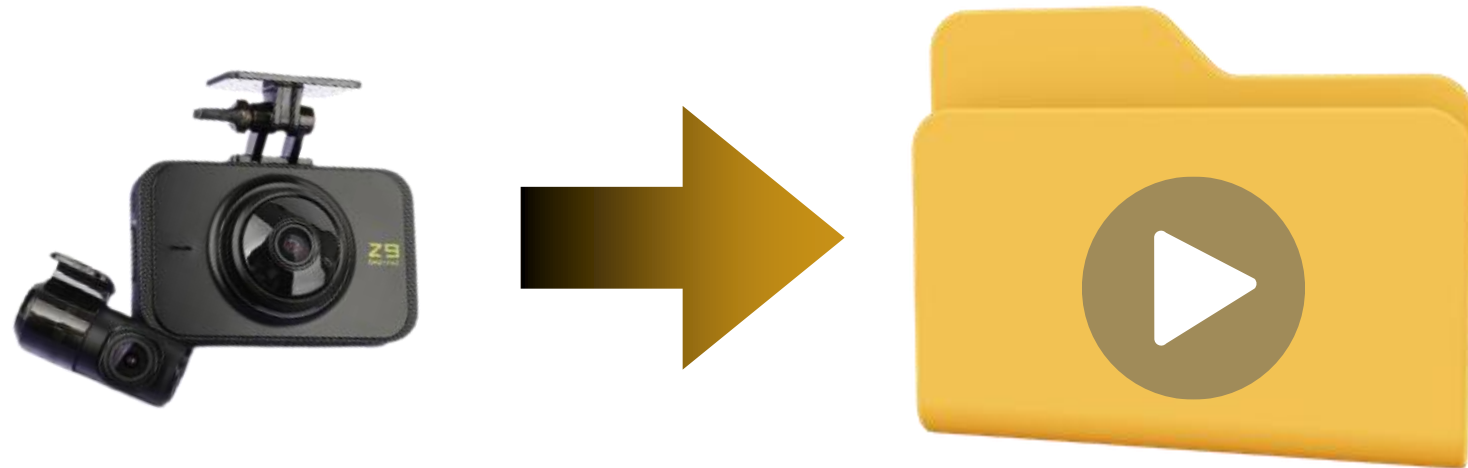
2







PASSIVE RECORDING SYSTEM



Passive Recording System
Records visual data
Used after an incident
No real-time feedback
No decision support.

REAL-TIME DECISION SUPPORT SYSTEM

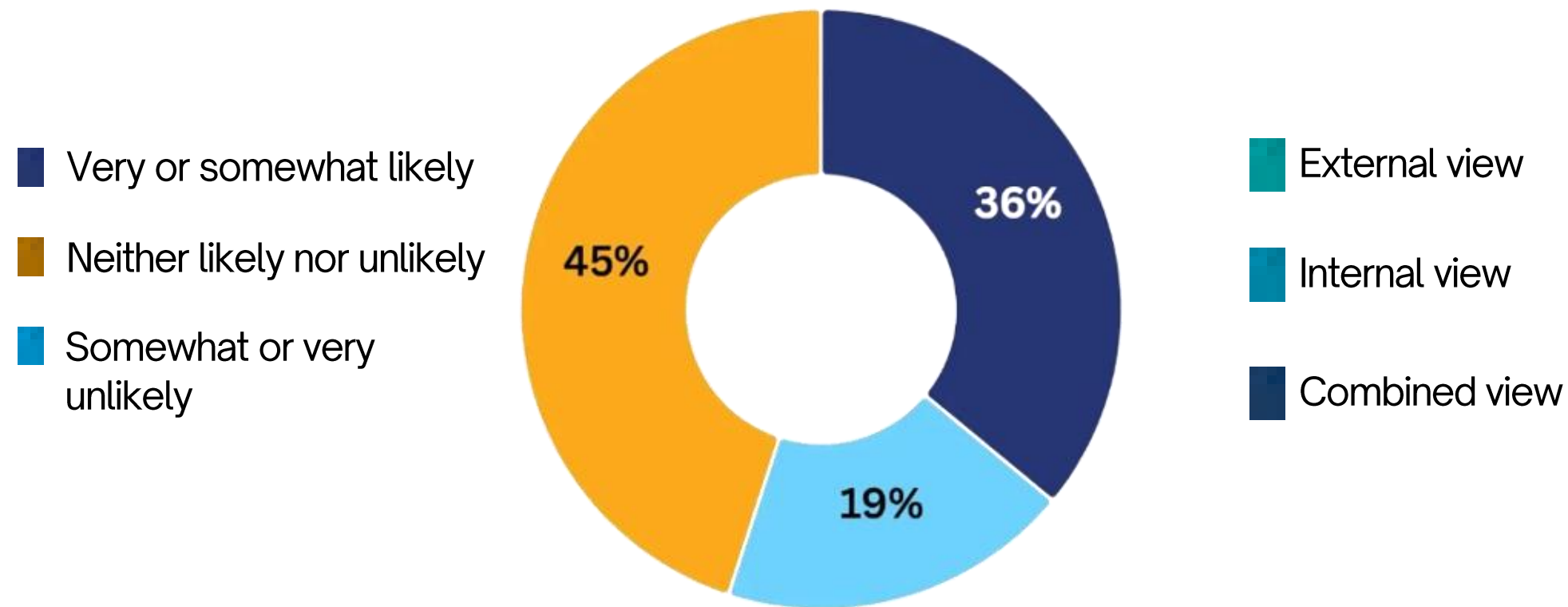


Real-time Decision Support System

Real-time spatial risk estimation
Detects unsafe space BEFORE entry
Instant GO / STOP decision cues
Human-in-the-loop assistance

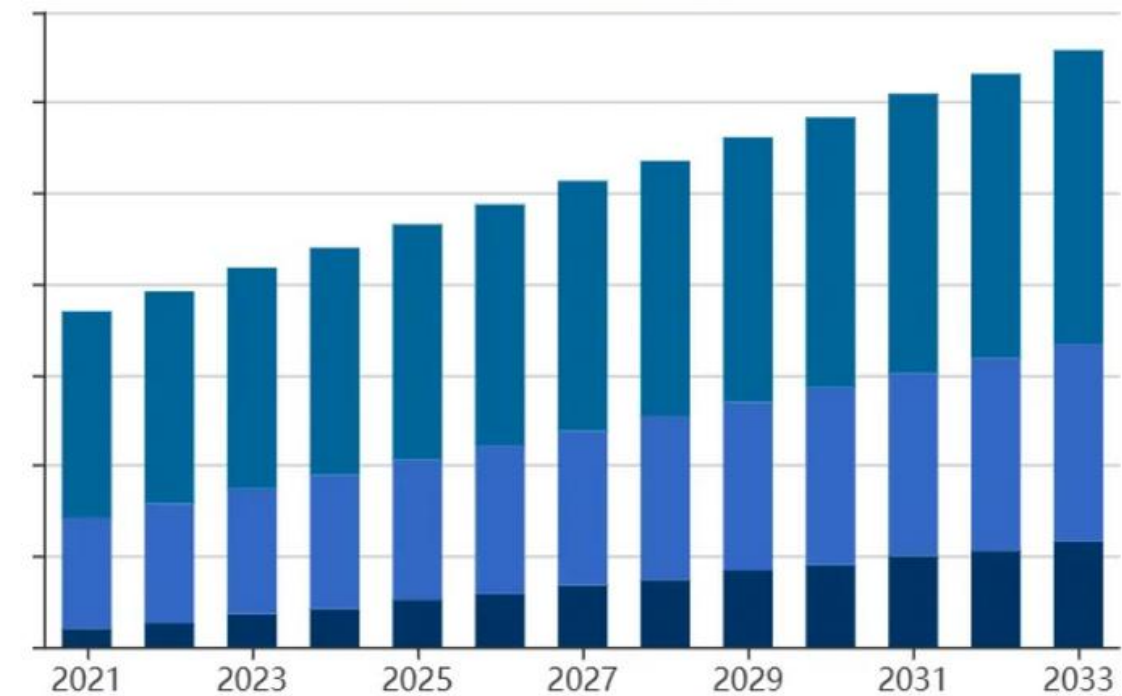
Hardware is replaceable. The decision-support algorithm is the core value

U.S. DASH CAM MARKET DEMAND IS INCREASING



AutoInsurance.com, Dash Cam Usage Report 2025

36% of non-users plan to buy within 1 year

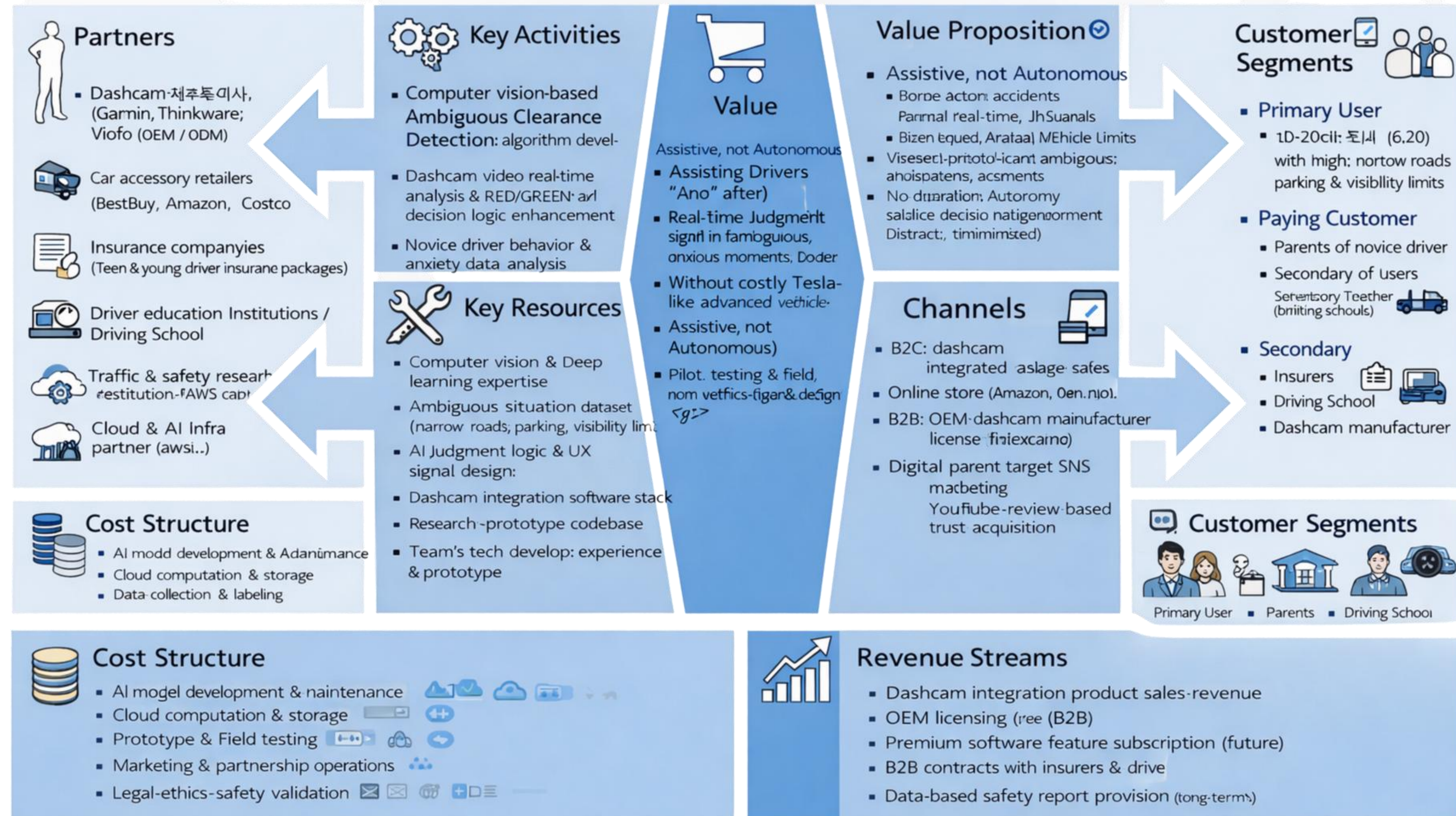


Cognitive Market Research, North America Automotive Dash Camera Market Report

North America dash cam market shows sustained growth through 2033

Business Model

Ambiguous Clearance Detection – AI Decision Support for Drivers



THANK YOU FOR LISTENING

Team. 11

